

Code smells and best practices for clean code

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With experience, you gain instincts that tell you when a piece of code is poorly written, hard to understand, or prone to generate bugs or runtime errors.

And then other people's code can start to **smell bad**.

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Code smells are heuristics that indicate when to refactor.

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- This is hard to read.
- This is hard to test.
- If I change this, things elsewhere break.
- I find myself changing this back and forth.



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- Extract duplicated code into a reusable function or class.

```
def chainMean(samples, parameter):  
    chain = samples[0][parameter].flatten()  
    return numpy.mean(chain)
```

Duplicated code — calls and tests

```
chainMean(samples, 'alpha')  
chainMean(samples, 'gamma')  
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```
def testChainMean(unittest.TestCase):  
    ...
```

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```
def integrand1(self, p):  
    return self.pdf(p, self.n, self.k) * self.prior1(p)  
def integrand2(self, p):  
    return self.pdf(p, self.n, self.k) * self.prior2(p)
```

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- Manage responsibilities
 - A class or method should generally have **one** responsibility.

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- Manage responsibilities
 - A class or method should generally have **one** responsibility.
 - Don't split a single responsibility among classes or methods.

- Frequently changing what a particular identifier refers to.

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Variable mutations — avoid rebinding

- Frequently changing what a particular identifier refers to.

```
from math import comb
```

```
def binomial_pdf(obj, p):  
    p = comb(obj.n, obj.k) * (p**obj.k) * ((1 - p)**(obj.n - obj.k))  
    return p
```



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```
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```
def binomial_pdf(obj, p):  
    n_win, n_lose = obj.get_k(), obj.get_n() - obj.get_k()  
    p_win, p_lose = p, 1 - p  
    out = comb(n_win + n_lose, n_win) * (p_win**n_win) * (p_lose**n_lose)  
    return out
```

Uncontrolled side effects

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Respect scope.

```
class myClass:  
    def __init__(self, X):  
        self.X = X
```

```
def getX(obj):  
    X = obj.X  
    obj.X = None  
    return X
```

```
n = myClass(5)
```

```
(n.X, getX(n), n.X)
```

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```
class Wallet:
    def __init__(self, balance):
        self.balance = balance

def peek_balance(wallet: Wallet) -> int:
    return wallet.balance

w = Wallet(5)
(w.balance, peek_balance(w), w.balance) # --> (5, 5, 5)
```



Data clumps

- Multiple related variables that are often passed together, indicating a need for a class.



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```
def show_fixation(width_px, height_px, distance_cm, gamma, bg_color):  
    ...
```

```
def run_trial(subject_id, width_px, height_px, distance_cm, gamma, bg_color):  
    show_fixation(width_px, height_px, distance_cm, gamma, bg_color)  
    # Same display settings travel together on every call: a clump.
```



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```

```
    ...
```

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```
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    # Same display settings travel together on every call: a clump.
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```
from dataclasses import dataclass
```

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def mean(xs, weights=None, ignore_nans=False, trim_fraction=0.0):  
    return sum(xs) / len(xs)
```

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```
def label_trial(rt_ms, correct, practice):  
    if practice:  
        if rt_ms < 150:  
            if not correct:  
                return "practice_fast_error"  
            return "practice_fast_correct"  
        if rt_ms > 3000:  
            if not correct:  
                return "practice_slow_error"  
            return "practice_slow_correct"
```

Too many parameters — crowded signature

- Makes calling and testing a function complicated

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```
def plot_experiment(  
    iv_data,  
    iv_names,  
    dv_data,  
    dv_names,  
    covariates,  
    covariate_names,  
    experiment_name,  
    aesthetic,  
):  
    ... # do stuff ...
```

Too many parameters — reshape responsibility

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```
class Experiment:
```

```
...    # Class to contain experiment data ...
```

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```
run_simulation_study(n_repetitions):  
    parameters          = sample_parameters(n_repetitions)  
    simulated_data       = simulate_data(parameters)  
    estimated_parameters = estimate_parameters(simulated_data)  
    return compare_parameters(  
        true_parameters      = parameters,  
        estimated_parameters = estimated_parameters  
    )
```



Excessively long or short identifiers

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compute_marginal_likelihood_from_binomial_data_and_priors(  
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)
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)  
  
ml(d1, pr, uc)
```



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Excessively long line of code — dense expression

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```
subtab = data.loc[subset].groupby(  
    factors  
)[value].agg(  
    [numpy.mean, numpy.std, len]  
)
```

Excessively long line of code — stepped layout

- Break up long lines into smaller, more manageable chunks

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```
statistics_list = [numpy.mean, numpy.std, len]
subset          = data.loc[subset]
grouped_subset  = subset.groupby(factors)
grouped_value   = grouped_subset[value]
aggregated_value = grouped_value.agg(statistics_list)
```

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- Practice **contractual programming**: document the **interface** carefully



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- **Large class:** a class that contains many unrelated methods.
- **Lazy class:** a class that does too little.
- **Feature envy:** a class that uses methods of another class excessively.
- **Excessive use of literals:** these should be coded as named constants, to improve readability and to avoid programming errors.

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- The names should clearly indicate the purpose and behavior of the code.
- Example:
 - Instead of using a variable name like `x`, use `customerName`
 - Instead of using a function name like `process`, use `calculateDiscount`



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Bad naming practice

```
lst = [1, 2, 3]
```

```
dct = {"a": 1, "b": 2}
```

```
def fnc(a, b):
```

```
    return a + b
```

```
class Cls:
```

```
    def __init__(self):
```

```
        self.x = None
```

```
        self.y = None
```

```
    def mtd(self):
```

```
        pass
```

Naming conventions — good practice

- Good naming practices make code more self-documenting and easier to maintain

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Good naming practice

```
numbers_list = [1, 2, 3]
```

```
data_dict = {"height": 1, "weight": 2}
```

```
def addNumbers(firstNumber, secondNumber):  
    return firstNumber + secondNumber
```

```
class DataProcessor:  
    def __init__(self):  
        self.current_data = None  
        self.previous_data = None  
    def process_data(self):  
        pass
```

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- Avoid long functions with multiple responsibilities.
- Example:
 - Instead of having a long function that retrieves data from a database, processes it, and sends an email all in one, have three separate functions: one for retrieving data, one for processing data, and one for sending emails.



```
def calculate_and_save_results(numbers):  
    result = 0  
    for number in numbers:  
        result += number  
    with open("result.txt", "w") as f:  
        f.write(str(result))  
    return result
```



Refactor functions — after

```
def calculate_sum(numbers):  
    result = 0  
    for number in numbers:  
        result += number  
    return result  
  
def save_to_file(result, file_path):  
    with open(file_path, "w") as f:  
        f.write(str(result))  
  
numbers = [1, 2, 3, 4]  
result = calculate_sum(numbers)  
save_to_file(result, "result.txt")
```

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 - Instead of writing a comment like “increment the value of x” before “x += 1”... don’t write that
- If you find you need to explain your code a lot, you might want to simplify the code rather than apologizing for it



Bad commenting practice

```
def calculate_sum(numbers):  
    # Calculating the sum of numbers  
    result = 0  
    for number in numbers:  
        result += number  
    # Return result  
    return result
```



```
def calculate_sum(numbers):  
    """  
    Calculate the sum of a list of numbers.  
  
    Arguments:  
    numbers -- list of numbers to be summed  
  
    Returns:  
    result -- the sum of the numbers  
    """  
    result = 0  
    for number in numbers:  
        result += number  
    return result
```



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 - Use a consistent indentation level throughout the code.
 - Use consistent capitalization for variables and functions.
 - Use consistent spacing and alignment.

```
def calculate_sum(numbers):  
    output_arg=0  
    for i in range(len(numbers)): output_arg=output_arg+numbers[i]  
    return output_arg
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- Use exceptions to indicate **what** has gone wrong.
- In general, do not try to fix a user's input error. Wrong input should give an informative error message to set the user straight. **Code should never quietly give a result that might not be what the user wanted.**

Good Error Handling Practice

✓ Error handling — good practice

Good Error Handling Practice

```
def centered_normpdf(x, sigma):  
    if sigma == 0:  
        raise ValueError("The standard deviation" +  
                           " must be non-zero.")  
    return (1 / (math.sqrt(2 * math.pi) * sigma)) * \  
           math.exp(-0.5 * (x / sigma) ** 2)  
  
print(centered_normpdf(0, 0))  
# ValueError: The standard deviation must be non-zero.
```

Inferior Error Handling Practice



Error handling — inferior practice

Inferior Error Handling Practice

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def centered_normpdf(x, s):  
    return (1 / (math.sqrt(2 * math.pi) * s)) * \  
        math.exp(-0.5 * (x / s) ** 2)  
  
print(centered_normpdf(0, 0))  
# ZeroDivisionError: float division by zero
```

Overly Defensive Error Handling Practice

Overly Defensive Error Handling Practice

```
def centered_normpdf(x, s):  
    s = max(0.0001, s)  
    return (1 / (math.sqrt(2 * math.pi) * s)) * \  
        math.exp(-0.5 * (x / s) ** 2)  
  
print(centered_normpdf(0, 0))  
# 3989.4228040143275
```

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 - A good test suite gives you confidence to make changes to the code without fear of breaking existing functionality.

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